Market Change, Assessment Systems, and Property-Tax Incidence: Evidence from Benton County's Calibrated Cost Governance

Full research paper (filled draft)

Prepared: January 13, 2026 (Updated: January 19, 2026)

Benton County Assessor's Office (draft for internal review)

Workfile/Version: Draft v0.3 (internal)

Executive summary (policy and practice)

- Property tax bills change for two distinct reasons: (1) taxable value (assessment) and (2) levy/rate decisions. Public frustration often attributes the full bill change to assessment alone.
- This report treats Benton County's calibrated cost system as a governed measurement system: diagnostics (level, uniformity, vertical equity) trigger bounded, versioned adjustments that must pass holdout validation.
- Using the provided 2022 parcel tax-bill extract (N=91,876), the median tax bill is $2,232 with an interquartile range of $683–$3,496; total billed taxes sum to $274,891,498.
- Using 2024 subset-level vertical equity diagnostics (35 subsets; 908 qualified sales), 40% of subsets show statistically significant price-related bias ($p<0.05$), identifying where regressivity risk concentrates.
- A counter-ready bill decomposition (value vs rate vs interaction) improves taxpayer communication and reduces appeals friction by showing what is (and is not) caused by assessment.
- Governance artifacts (gate-based targeting, documented change lists, before/after validation packets) provide hearing-ready defensibility while maintaining a sales-chasing firewall.
- Policy implication: levy-limit mechanics and inflation interact with local service demands; assessment administration can improve equity and stability, but it cannot substitute for policy decisions about revenue growth and allocation.

How to read this draft: Sections 3–6 define the data, diagnostics, and decomposition. Sections 7–8 describe the governance controls and equity lens. Appendix A crosswalks evidence bundles used in validation and public communication.

Abstract

This paper connects neighborhood change, assessment performance, and tax-bill outcomes using Benton County's market-calibrated cost approach and governance system. We integrate (i) ratio-study diagnostics (level, uniformity, and vertical equity), (ii) a decomposition of tax-bill change into value and rate components, and (iii) a fail-closed calibration workflow that documents changes via versioned tables/curves and validates deployments with holdout testing. Using the provided Benton County tax-bill extract for 2022 (N=91,876 parcels), the median parcel tax bill is $2,232 (IQR $683–$3,496) and total billed taxes sum to $274,891,498. Separately, using the 2024 "all subsets" tilt diagnostics (35 subsets; 908 qualified sales), 40% of subsets show statistically significant price-related differential ($p<0.05$) with a median log-price tilt coefficient of -0.459, indicating where vertical equity stress concentrates. We show how Benton's governance artifacts—gate-based subset targeting, documented change lists, and before/after validation packets—translate these diagnostics into auditable adjustments while maintaining a sales-chasing firewall. Finally, we outline an equity-lens reporting approach that can incorporate area-level demographics descriptively (with suppression) without using protected-class variables as valuation inputs.

Keywords

property tax; mass appraisal; vertical equity; reassessment; neighborhood change; tax incidence; governance; sales chasing; calibrated cost approach

1. Introduction

Property tax fairness debates often collapse distinct mechanisms into a single complaint: “the assessor raised my taxes.” In practice,
a taxpayer’s bill can change because taxable value changes, because levy rates change, or because both change simultaneously.
At the same time, housing market dynamics—turnover, redevelopment, rapid price growth, and shifting demand—can stress any mass appraisal
system and create systematic patterns in assessment ratios and vertical equity metrics. Those patterns can be perceived as unfairness
between neighborhoods or price tiers, and they can intensify skepticism about practices such as “sales chasing.”

This paper frames the problem as an engineering-and-governance question: how should an assessor’s office measure market change,
detect where performance degrades, and then deploy calibrations in a way that is auditable, repeatable, and explicitly fail-closed?
We develop that frame using Benton County’s market-calibrated cost approach and a set of standardized governance artifacts
(Top-subset targeting, documented change lists, before/after validation packets, and a sales-chasing firewall attestation).

We also align our work with two complementary research traditions. First, neighborhood-change research emphasizes how dynamics such as
gentrification and turnover can interact with assessment systems to produce distributional consequences, including racial inequality in
property tax burdens. Second, reassessment research emphasizes that property tax incidence depends critically on assessment accuracy:
even an otherwise proportional (ad valorem) tax becomes regressive or unfair when assessments deviate systematically across value tiers
or neighborhoods.

1.1 Research questions

1. RQ1: Where does assessment performance degrade when neighborhoods change quickly (turnover, price growth, new construction, dispersion)?
2. RQ2: Do Benton’s calibration gates and versioned interventions reduce vertical equity defects and instability over time?
3. RQ3: How much of observed tax-bill change is driven by taxable value versus levy/rate (and their interaction), and how does that vary by value tier and exemption status?
4. RQ4: How can an assessor report distributional patterns with an equity lens—potentially including area-level demographics descriptively—without using protected-class variables as valuation inputs?

1.2 Contributions

(1) A tax-bill decomposition that separates bill changes into value and rate components, suitable for public counter use and for
distributional analysis (Exhibit/Appendix TAX-1). (2) A neighborhood-change and QA clustering framework that produces triage-ready
hotspots without embedding demographics into valuation (Exhibit/Appendix NC-1). (3) A governance evidence bundle that operationalizes
a sales-chasing firewall through scope locks, versioned change logs, and holdout validation (Exhibit/Appendix SC-1, plus D/E/F).
(4) A matched-pair (stratified matching) audit that enables apples-to-apples performance comparisons across groups/areas after
holding observable property characteristics constant (Exhibit/Appendix MP-1).

2. Institutional Background

2.1 Benton's market-calibrated cost system (overview)

Benton's internal one-page description of the system summarizes the workflow as follows (excerpted from the internal one-pager):

Figure 1. Benton market-calibrated cost governance (simplified).

Inputs -> Calibration -> Quality assurance -> Governance

- Inputs: Verified arms-length sales; property characteristics; local construction cost inputs.
- Calibration: Estimate percent-of-base curves and local cost tables from sales evidence; retain only characteristics with stable price signal.
- Quality assurance: Run ratio-study metrics (level/uniformity) and vertical equity diagnostics (PRD/PRB); segment by class/neighborhood/tier.
- Governance: Deploy only versioned, documented changes that pass holdout tests; maintain change logs and peer review sign-off.

Sales-chasing firewall controls (operational).

Operational artifacts (examples in workbooks):

- SalesDashboard_Ratio Sheets.xlsx (raw sales + charts)
- Chris Dashboard Test.xlsx (all parcels; ratio sheet; analytics/statistics)
- Revaluation trend sheets (neighborhood/subdivision ratio trends)

Control	What it blocks	How it is evidenced
Temporal separation	Adjusting to individual sales	Defined study period; sales cutoff lock; effective date and

		roll date documented.
Holdout / out-of-sample validation	Overfitting the sold set	Holdout sample maintained; pre/post metrics reported on holdout and unsold proxies where available.
Model-first, sale-second changes	Manual tuning to “hit” a sale	Changes applied via tables/curves/matrices rather than parcel edits; exceptions are rare and logged.
Stratified equity tests	Regressivity/progressivity	PRD and PRB by tiers/segments compared before/after; neutrality required or explicit policy rationale documented.
Versioned change governance	Unreviewed tweaks	Version tables; change memos; peer review sign-off; reproducible artifacts for audit/appeal.

2.2 Why governance matters: sales chasing versus controlled calibration

A core concern in public discussions is “sales chasing,” typically understood as adjusting assessed values on sold parcels to match
sales while leaving unsold parcels lagging. Benton’s governance model is designed to avoid that failure mode by enforcing:
(i) scope locks (sales cutoff and model lock), (ii) segment-level changes applied through tables/curves/matrices rather than parcel-by-parcel overrides,
(iii) holdout validation with fail-closed deployment rules, and (iv) a complete versioned change log tied to objective diagnostics.
These controls make the calibration process auditable and reproducible, and they produce a record that can be reviewed internally,
by auditors, or in an appeals context.

2.3 Key diagnostic definitions

We use standard ratio-study and vertical equity diagnostics that are widely referenced in professional practice:
median assessment ratio (level), coefficient of dispersion (uniformity), and price-related differential metrics (e.g., PRD/PRB).
To diagnose vertical equity more directly, we also use tier plots and a controlled “tilt” regression

that estimates how assessment
error varies with price (reported as a slope on log price with a p-value). In this paper, these diagnostics are treated as measurement
outputs that trigger investigation and governance steps; they are not used to target outcomes for any protected class.

3. Related Literature

3.1 Neighborhood change and racial inequality in property taxes

The RAND dissertation "Neighborhood Change, Housing Market Dynamics, and Racial Inequality in Property Taxes" situates property tax
inequality within persistent racial wealth inequality and emphasizes the role of neighborhood change, gentrification, and sales volume.
Its abstract highlights mechanisms through which local assessment practices, market dynamics, and neighborhood amenities can interact,
and argues for routine, high-quality reassessment as part of the policy response. We summarize the argument here rather than reproducing extended text excerpts.

O'Hollaren (RAND) documents how neighborhood market dynamics and reassessment practices can interact to produce systematic assessment gaps. A central empirical pattern in that work is that jurisdictions/areas with higher sales volume and market transparency tend to show smaller gaps, supporting a practice implication: routine ratio-study monitoring and governed recalibration are necessary conditions for equitable, defensible assessments.

3.2 Reassessment and the importance of assessment accuracy

Christopher Berry's "Reassessing the Property Tax" emphasizes that the property tax is a major revenue source and is often viewed as
progressive because it taxes property wealth. However, Berry argues that the practical incidence of the tax hinges on assessment accuracy:
systematic assessment errors can make the tax regressive (e.g., low-priced properties over-assessed relative to market value) and can
create inequities even when statutory tax rates are uniform.

In Berry's framing, even a uniform statutory rate can yield regressive outcomes if assessment errors are systematically larger for lower-priced homes. The policy lever is not only the rate structure, but also reassessment frequency, data quality, and transparent quality control that keeps assessment ratios stable across value tiers.

3.3 How this paper differs

Most academic work observes outcomes (effective tax rates, assessment ratios, spatial inequality) but has limited visibility into the
internal calibration governance that produces those outcomes. Benton's contribution is an

“audit trail” approach: diagnostics are connected
to explicit gate triggers, versioned changes, and holdout validation. That structure allows research-style claims to be paired with
hearing-ready evidence about what changed, why, and how deployment was validated.

4. Data

4.1 Parcel-level tax-bill extract (2022)

We use a parcel-level extract with taxable value, total tax rate (in mills), and computed total tax for tax year 2022
(N=91,876). The median total rate is 8.98 mills. In this extract,
4.0% of parcels are flagged as having at least one exemption indicator. Because this dataset was provided
as an extract for analysis, it should be treated as “as-delivered” for this draft; final publication should document the extraction date,
filters, and any exclusions (e.g., non-taxable parcels).

Table 1. Tax-bill summary statistics (2022 extract)

Parcels (N)	Total tax ($)	Median tax ($)	P25 tax ($)	P75 tax ($)	Exempt share
91,876	274,891,498	2,232	683	3,496	4.0%

4.2 Tax distribution by assessed-value tier (2022)

The tier distribution file summarizes parcel counts and selected tax percentiles across five value tiers (P0–25, P25–50, P50–75, P75–90, P90–100).

Table 2. Tax distribution by tier (2022)

tier	parcels	median_assessed	median_tax	p25_tax	p75_tax	p90_tax
P0-25	21614	16,335.00	177.53	10.01	553.16	962.89
P25-50	21619	210,690.00	1,837.60	1,512.38	2,243.56	2,583.37
P50-75	21609	307,690.00	2,829.80	2,402.45	3,327.17	3,860.37
P75-90	12968	424,495.00	4,128.61	3,485.09	4,815.31	5,427.26
P90-100	8646	649,410.00	6,490.87	5,411.26	8,708.09	17,257.57

4.3 Exempt versus non-exempt comparison (2022)

A second summary file compares exempt-flagged parcels versus non-exempt parcels. Differences reflect both value differences and statutory/program eligibility.

Table 3. Exempt vs non-exempt summary (as provided)

is_exempt_any	parcels	median_assessed	median_tax

False	82857	269,390.00	2,433.05
True	3599	25,060.00	128.90

4.4 Vertical equity / tilt diagnostics (2024 all-subsets report)

We use an “all subsets” tilt diagnostic file (35 subsets; 908 sales observations) that reports standard ratio-study metrics
(median ratio, COD, PRD) and a log-price tilt coefficient with p-values. These subset-level results are designed for internal triage:
they identify which market slices exhibit systematic vertical equity stress and therefore merit targeted calibration review.

Table 4. Tilt diagnostics summary (2024 subsets)

Subsets	Sales observations	Share with tilt p<0.05	Median tilt beta (ln price)
35	908	40.0%	-0.459

4.5 Demographic context (equity lens) — not included in current extract

The RAND/Chicago-style framing motivates describing whether property tax burdens or assessment performance differ across areas with
different demographic compositions. That requires an area-level demographic dataset (e.g., ACS tract or block-group estimates) joined
to parcels by geography. Demographic inputs were not included in the provided extract set for this draft. In a final version, any
demographic overlays should be: (i) aggregated to an appropriate geography (tract/block group), (ii) suppressed for small-cell privacy,
and (iii) treated as descriptive context only—not as valuation inputs.

5. Methods

5.1 Ratio-study and vertical equity diagnostics

We treat ratio-study metrics as measurement outputs that provide evidence about performance (level, uniformity, and vertical equity).
In operational use, these metrics are calculated within defined segments/subsets to detect whether performance issues are local or systemic.

Level. We summarize central tendency using the median assessment ratio. Uniformity. We summarize dispersion using the coefficient of
dispersion (COD). Vertical equity. We evaluate whether ratios vary systematically with value using PRD/PRB-style measures and by

estimating a controlled “tilt” coefficient from a regression of assessment error on log price within each subset.

5.2 Tax-bill decomposition (TAX-1 identity)

For a parcel i, with taxable value V and total tax rate R (mills), the tax bill is Tax = V·R/1000. Between Year 1 and Year 2 we can
decompose the change in tax into a value component, a rate component, and an interaction term:

$\Delta Tax_i = Tax_i,2 - Tax_i,1$
$ValueEffect_i = (V_i,2 - V_i,1) \cdot R_i,1 / 1000$
$RateEffect_i = V_i,1 \cdot (R_i,2 - R_i,1) / 1000$
$CrossTerm_i = (V_i,2 - V_i,1) \cdot (R_i,2 - R_i,1) / 1000$

This identity is the foundation for counter-ready explanations (“your bill changed because X% value, Y% rate”) and for distributional
summaries by tier, exemption status, or geography.

5.3 Governance workflow and the sales-chasing firewall (SC-1; D/E/F)

Benton’s workflow is designed to convert diagnostics into controlled calibration changes:

Step 1 — Diagnose. Identify failing subsets using objective metrics and coverage notes (e.g., tier drift, |tilt|, PRD deviation).
Step 2 — Target. Prioritize a limited number of subsets (“Top 5”) for review to focus effort where risk is highest.
Step 3 — Hypothesize. Document likely causes (e.g., base cost tables vs depreciation/effective age mapping vs condition/quality stratification).
Step 4 — Change via tables/curves/matrices. Implement changes at segment/subset scope (not parcel-by-parcel), version tagged.
Step 5 — Validate. Produce before/after packets including in-sample and holdout results. Fail-closed if holdout gates fail.
Step 6 — Archive and attest. Retain workfiles and generate a firewall attestation packet (scope locks, change logs, holdout pass).

This structure provides both (i) technical defensibility and (ii) an administrative record that is legible to non-technical reviewers.

5.4 Matched-pair (stratified matching) audit (MP-1)

To examine whether assessment performance differs across two groups after controlling for observable property characteristics,
we construct match “cells” defined by bands of square footage and age plus categorical condition, quality, and class (and optionally
a neighborhood segment). Within each cell, we compute mean performance for Group A and Group B and then average the within-cell

differences using pooled weights (nA+nB). Cells without overlap (nA=0 or nB=0) are excluded and reported as “unmatched,” which is itself
a diagnostic indicator of structural differences between groups.

5.5 Econometric extensions (recommended for publication)

To more closely match the style of neighborhood-change research, a final publication can estimate neighborhood-year panel models in which
vertical equity defects (tier gap, tilt magnitude, PRD deviation) are outcomes, neighborhood change indicators are key predictors, and
neighborhood and year fixed effects control for time-invariant differences and common shocks. An event-study design can be layered on
to evaluate changes in defects around documented calibration deployments (version-tagged interventions). These extensions require a
multi-year panel of diagnostics and neighborhood indicators; only partial year coverage is included in the current extract set.

6. Results

6.1 Vertical equity stress: tilt diagnostics across subsets (2024)

Across the 35 evaluated subsets (908 sales), 40.0% show statistically significant log-price tilt at the 5% level. The median tilt
coefficient is -0.459. A negative tilt coefficient indicates that ratios decline with price within the subset
(i.e., higher-priced properties tend to be under-assessed relative to lower-priced properties, conditional on the subset definition).
These results support the operational premise that vertical equity stress is not uniform: it concentrates in specific market slices and
is therefore best managed via targeted calibration rather than broad, undifferentiated changes.

Table 5. Top 10 subsets by |tilt| (from 2024 all-subsets diagnostics)

subset	n_sales	median_ratio	COD	PRD	tilt_beta_lnprice	tilt_pvalue	median_price
211.0000	20.0000	1.0197	7.9845	0.9988	-0.8088	0.0096	594,950.0000
109.0000	34.0000	0.9108	6.5897	1.0066	-0.6901	0.0001	377,500.0000
146.0000	16.0000	0.9113	10.0091	1.0169	-0.6103	0.0192	435,250.0000
110.0000	63.0000	0.9562	5.7942	1.0028	-0.5520	0.0000	435,000.0000
107.0000	50.0000	0.9062	7.6646	1.0000	-0.5272	0.0000	382,450.0000
532.0000	42.0000	0.9805	3.3546	1.0014	-0.5075	0.0000	394,994.

							5000
105.0000	46.0000	0.8949	10.5883	1.0034	-0.4837	0.0011	306,950.0000
209.0000	26.0000	0.9125	6.4986	1.0034	-0.4686	0.0002	519,000.0000
111.0000	53.0000	0.9457	5.9420	1.0089	-0.4663	0.0000	484,900.0000
212.0000	37.0000	0.9662	4.2383	1.0081	-0.4527	0.0004	563,514.0000

6.2 Gate-based targeting: Top subsets and recommended calibration focus (Exhibit D/E)

Benton's governance artifacts translate the diagnostics into a bounded review queue. The Top-5 targeting exhibit provides, for each
priority subset, a severity label, the tilt magnitude and significance, supporting distribution descriptors (age/sqft/price IQR),
and a narrative hypothesis about likely drivers. The companion change-list exhibit documents recommended matrix/table identifiers and
a rationale tied to the failed gates. Together, these exhibits show how diagnostics become structured action: a reviewer can see (i)
why a subset was flagged, (ii) what levers are implicated, and (iii) what documentation is required before deployment.

Table 6. Calibration target list (as provided in Exhibit D)

subset	**severity**	**n_sales**	**median_ratio**	**COD**	**PRD**	**tilt_beta_lnprice**	**tilt_pvalue**	**median_price**	**focus**	**hypothesis**	**recommended_matrix_ids**
211	Severe	20	1.0197	7.9845	0.9988	-0.8088	0.0096	594,950.0000	Finish/remodel proxies, Segmentation review	Condition mostly one code -> condition proxy may be too coarse;	1598, 3249, 3251, 3253, 3260, 3262, 3263, 3265, 3266, 3267, 3296, 3300, 3320, 3789,

										unobserved finish/remodel likely.	3790, 3791, 3792, 3793, 3794, 3805, 3806, 3807, 3808, 3809, 3810, 3811, 3812, 3813, 3814, 3815, 3816, 3817, 3818, 3823
109	Severe	34	0.9108	6.5897	1.0066	-0.6901	0.0001	377,500.0000	Finish/remodel proxies	Condition mostly one code -> condition proxy may be too coarse; unobserved finish/remodel likely.	1598, 3249, 3251, 3253, 3260, 3262, 3263, 3265, 3266, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3807, 3808, 3809, 3810, 3811, 3812, 3813, 3814, 3823
146	Sever	16	0.911	10.00	1.016	-0.61	0.019	435,2	Segm	Likely	3789,

	e		3	91	9	03	2	50.0000	entation + outlier/filters	mix of unobserved quality and segmentation heterogeneity; prioritize feature proxies + segment split review.	3790, 3791, 3792, 3793, 3794, 3823
110	High	63	0.9562	5.7942	1.0028	-0.5520	0.0000	435,000.000	Finish/remodel proxies	Condition mostly one code -> condition proxy may be too coarse; unobserved finish/remodel likely.	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823
107	High	50	0.906	7.664	1.000	-0.52	0.000	382,4	Finis	Likely	1598,

			2	6	0	72	0	50.0000	h/remodel proxies, Segmentation review	mix of unobserved quality and segmentation heterogeneity; prioritize feature proxies + segment split review.	3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823
532	High	42	0.9805	3.3546	1.0014	-0.5075	0.0000	394,994.5000	Finish/remodel proxies	Condition not differentiating (single condition code) -> finish/remodel intensity likely unobserved.	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823, 3846
105	High	46	0.8949	10.5883	1.0034	-0.4837	0.0011	306,950.00	Segmenta	Likely mix	3789, 3790,

								00	tion + outlier/filters	of unobserved quality and segmentation heterogeneity; prioritize feature proxies + segment split review.	3791, 3792, 3793, 3794, 3823
209	High	26	0.9125	6.4986	1.0034	-0.4686	0.0002	519,000.0000	Segmentation review	Likely mix of unobserved quality and segmentation heterogeneity; prioritize feature proxies + segment split	3789, 3790, 3791, 3792, 3793, 3794, 3805, 3806, 3823

										review.	
111	High	53	0.9457	5.9420	1.0089	-0.4663	0.0000	484,900.0000	Finish/remodel proxies, Segmentation review	Condition not differentiating (single condition code) -> finish/remodel intensity likely unobserved.	1598, 3262, 3267, 3296, 3300, 3320, 3789, 3790, 3791, 3792, 3793, 3794, 3815, 3816, 3817, 3818, 3823
212	High	37	0.9662	4.2383	1.0081	-0.4527	0.0004	563,514.0000	Finish/remodel proxies, Segmentation review	Condition not differentiating (single condition code) -> finish/remodel intensity likely unobserved.	1598, 3249, 3251, 3253, 3260, 3262, 3263, 3265, 3266, 3267, 3296, 3300, 3320, 3789, 3790, 3791, 3792, 3793, 3794, 3805, 3806, 3807, 3808, 3809, 3810,

											3811, 3812, 3813, 3814, 3815, 3816, 3817, 3818, 3823, 3846
210	High	29	0.9542	5.4472	0.9987	-0.4260	0.0080	482,000.0000	Finish/remodel proxies	Condition mostly one code -> condition proxy may be too coarse; unobserved finish/remodel likely.	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823
145	High	23	0.9428	6.7296	1.0118	-0.4136	0.0066	350,000.0000	Finish/remodel proxies, Segmentation review	Likely mix of unobserved quality and segmentation heterogeneity; priori	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823

										tize feature proxies + segment split review.	
247	Moderate	32	0.8881	5.3403	1.0009	-0.2934	0.0374	431,500.0000	Finish/remodel proxies, Segmentation review	Likely mix of unobserved quality and segmentation heterogeneity; prioritize feature proxies + segment split review.	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823
112	Mild	184	0.9613	4.2238	1.0026	-0.1902	0.0000	494,950.0000	Finish/remodel proxies	Condition not differentiating (single condition code) ->	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823

										finish /remodel intensity likely unobserved.	

Table 7. Top-5 change list (as provided in Exhibit E)

subset	severity	n_sales	median_ratio	COD	PRD	tilt_beta_lnprice	tilt_pvalue	recommended_matrix_ids
111	High	53	0.9457	5.9420	1.0089	-0.4663	0.0000	1598, 3262, 3267, 3296, 3300, 3320, 3789, 3790, 3791, 3792, 3793, 3794, 3815, 3816, 3817, 3818, 3823
110	High	63	0.9562	5.7942	1.0028	-0.5520	0.0000	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823

107	High	50	0.9062	7.6646	1.0000	-0.5272	0.0000	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823
532	High	42	0.9805	3.3546	1.0014	-0.5075	0.0000	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823, 3846
112	Mild	184	0.9613	4.2238	1.0026	-0.1902	0.0000	1598, 3262, 3267, 3296, 3300, 3789, 3790, 3791, 3792, 3793, 3794, 3823

6.3 Tax incidence and distribution: tier and exemption patterns (2022)

The 2022 parcel tax extract shows substantial dispersion in tax bills. The median bill is $2,232, with an interquartile range of $683–$3,496. Tier summaries show that median tax bills increase sharply across assessed-value tiers, as expected in an ad valorem system. Exempt-flagged parcels have substantially lower
median assessed values and lower median tax bills in the provided summary.
These descriptive distributions provide the baseline needed for the more important

decomposition question in taxpayer communication:
how much of a given bill change is value-driven versus levy/rate-driven. In this draft, we provide the decomposition identity and the
counter-ready workflow (TAX-1); a full year-to-year decomposition requires multi-year taxable value and rate series at parcel-year level.

6.4 Linking neighborhood change to QA risk (NC-1) — framework

The neighborhood-change + QA clustering pack (NC-1) is designed to connect market dynamics to where QA issues cluster. It constructs:
(1) a ChangeIndex from turnover, growth, new construction share, and dispersion; (2) a QARiskIndex from tier gaps, tilt magnitude, PRD
deviations, and holdout failures; and (3) a PriorityScore to triage neighborhoods for review. The NC-1 runner and appendix were produced
as part of this project; however, neighborhood-level indicator inputs were not included in the extract set provided for this filled paper.
In a final publication, NC-1 results would be presented as hotspot maps and ranked tables, then linked back to the calibration changes
and validation results documented in D/E/F.

6.5 Matched-pair equity audit (MP-1) — framework

The matched-pair audit (MP-1) answers a common high-stakes question: “If two homes are the same in observable characteristics, do they
experience different assessment performance depending on where they are?” The MP-1 runner constructs match cells using sqft/age bands and
condition/quality/class, computes within-cell differences between two groups, and aggregates those differences with pooled weights.
This tool is explicitly USPAP-safe as a descriptive QA audit: it does not use protected-class variables as valuation inputs and does not
claim causal mechanisms. In this filled paper, MP-1 is presented as the recommended deep-dive method for equity-lens analyses after the
basic tier and subset diagnostics have flagged potential issues.

7. Robustness and Sensitivity

To align with best practice and the tone of the academic literature, a final publication should include robustness checks in three domains:

(1) Metric sensitivity: alternative tier definitions (quantiles vs fixed cutpoints), log-error metrics, mean vs median performance measures,
and alternative vertical equity statistics. (2) Sample sensitivity: alternative sales qualification filters, thin-market pooling rules, and
placebo time windows. (3) Reporting sensitivity: suppression thresholds for small cells,

alternative geographic aggregations, and disclosure
controls for demographic overlays.

The governance artifacts (D/E/F and SC-1) naturally support robustness because each calibration change is versioned and tied to explicit
gates; alternative specifications can be evaluated and documented within the same evidence bundle.

8. Discussion

8.1 Connecting the research framing to operational governance

The RAND neighborhood-change framing emphasizes that housing dynamics can produce unequal burdens, including along racial lines, through
their interaction with assessment systems. The Berry reassessment framing emphasizes that assessment accuracy is the hinge: systematic errors
transform the incidence of the property tax, potentially making it regressive or uneven.

Benton's approach operationalizes both insights. First, it treats market change as a measurable stressor and organizes diagnostics to show where
performance degrades (subsets, tiers, and ultimately neighborhoods via NC-1). Second, it treats accuracy and vertical equity as governed outcomes:
diagnostics trigger a bounded set of interventions that must pass holdout validation and produce an audit trail. This is a practical blueprint
for translating research concerns into defensible administration.

8.2 Equity lens without protected-class valuation inputs

A central design choice is separating "measurement" from "modeling." Protected-class variables should not be used as valuation inputs. However,
equity concerns can and should be addressed through descriptive measurement: comparing distributions of bill changes and assessment performance
across geographies, tiers, and matched property profiles, and then using those measurements to prioritize review. When demographic context is used,
it should be aggregated (e.g., tract/block group), suppressed for privacy, and interpreted cautiously (descriptive, not causal).

8.3 Practical implications for taxpayer communication

The largest communication failure in property tax administration is explaining tax bills as if they were caused only by assessment.
The TAX-1 decomposition provides a direct remedy: it separates the value-driven component of a bill change from the levy/rate-driven component.
A counter-facing workflow should begin with decomposition, then (if needed) proceed to performance diagnostics (tier plots, subset metrics),

and only then to governance evidence (what was changed, why, and how it was validated). This sequencing reduces conflict and increases trust.

9. Conclusion

This paper provides a research-style framework grounded in Benton County's operational practice. We connect neighborhood change and housing market dynamics to assessment performance and property tax outcomes using a measurement-first approach: ratio-study metrics, vertical equity diagnostics, and tax-bill decomposition. We then show how Benton's calibrated cost governance converts diagnostics into auditable, versioned changes validated by holdout testing under a sales-chasing firewall. The result is a blueprint for counties that want both academic-grade measurement and hearing-ready defensibility: measure what matters, change only through governed levers, validate before deployment, and communicate tax outcomes by separating value effects from rate effects.

References

- O'Hollaren, K. (2025). Neighborhood Change, Housing Market Dynamics, and Racial Inequality in Property Taxes (RGSD-A4498-1). RAND Corporation.
- Berry, C. R. (2021). Reassessing the Property Tax (working paper; draft March 2021). University of Chicago Harris School of Public Policy.
- Benton County Assessor's Office. (2026). Benton's Market-Calibrated Cost System: market-calibrated cost workflow and governance (internal memorandum).
- Benton County Assessor's Office. (2026). Evidence bundle suite (TAX-1, NC-1, SC-1, MP-1; Exhibits D/E/F; standards appendices): runner outputs and validation packets (internal workfile).

Appendix A. Evidence bundle crosswalk (project artifacts)

A1. TAX-1 — Tax Bill Decomposition Runner and Pack: counter-ready explanations of bill changes (value vs rate vs interaction) and tier summaries.
A2. NC-1 — Neighborhood Change + QA Cluster Runner and Pack: ChangeIndex, QARiskIndex, and PriorityScore; map-export table (inputs required for final results).
A3. SC-1 — Sales Chasing Firewall Runner and Pack: timing locks, change logs, holdout summaries, and paste-ready attestation language.
A4. MP-1 — Matched-Pair Audit Runner and Pack: stratified matching on observable characteristics; standardized gap and unmatched diagnostics.
A5. Exhibit D — Top calibration targets and permanent gate spec (as provided in CSV/XLSX/PDF).
A6. Exhibit E — Change list and memo appendix (as provided in CSV/XLSX/PDF).
A7. Exhibit F — Before/After validation packet template (pre-wired for push-button workflow).
A8. STAND appendices — standards mapping and calibration memo shell.